Paper questions E.3: Radioactive decay [134 marks]

1. [Maximum mark: 1]

Two radioactive samples X and Y have the same half-life. Initially the ratio $\frac{\text{activity of X}}{\text{activity of Y}}$ is 4.

What is this ratio after 2 half-lives?

A. $\frac{1}{2}$

B. 1

C. 2

D. 4 [1]

Markscheme
D

2. [Maximum mark: 1]

Three statements about a nuclear fission reactor are:

I. The heat exchanger transfers energy from the fuel rods to the moderator.
II. The control rods must be good absorbers of neutrons.
III. The moderator must slow neutrons down.

Which statements about the reactor are correct?

A. I and II only

B. I and III only

C. II and III only

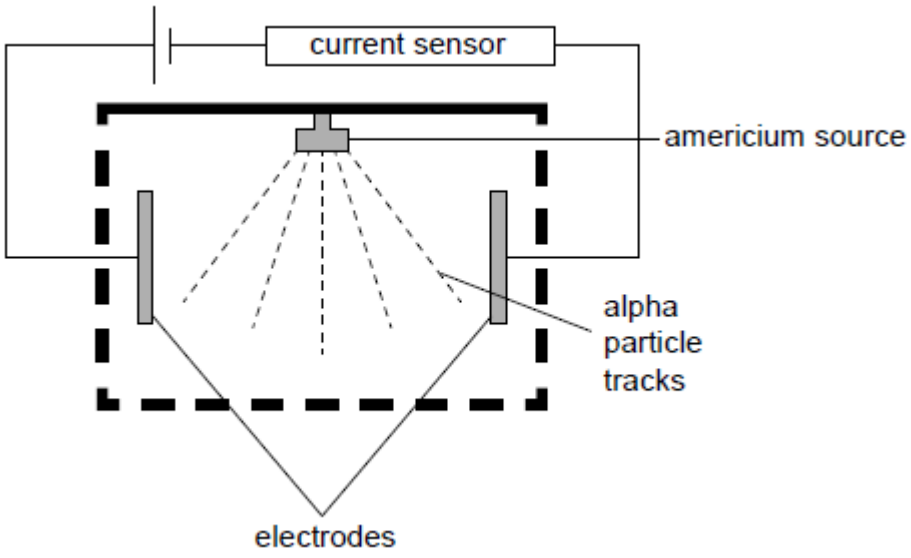
D. I, II and III [1]

Markscheme
C

3. [Maximum mark: 6]

A smoke detector uses the radioactive nuclide americium-241.

The americium is contained in a chamber that is open to the air. There are two electrodes in the chamber that are connected to a power supply and a current sensor.



Americium-241 emits alpha particles that ionize the air in the chamber. Each ionization forms one positive ion and one electron; these are called an ion pair. The electrons and the positive ions move towards the electrodes and the sensor detects a current in the air.

When smoke enters the chamber, fewer ion pairs are formed and the current in the sensor decreases, sounding an alarm.

The chamber is 0.10 m in each dimension.

(a) A nucleus of americium-241 has 146 neutrons. This nuclide decays to neptunium through alpha emission.

Complete the nuclear equation for this decay.

$$__\text{Am} \rightarrow __\text{Np} + __\alpha$$

[2]

Markscheme

$^{241}_{95}\text{Am}$ ✓

$^{237}_{93}\text{Np} + ^{4}_{2}\alpha$ ✓

(b) Outline why the radioactive source is safe for use in a house. [1]

Markscheme

Alpha particles only travel a few cm in air / penetration of alpha particles is poor (and will not escape the chamber) ✓

OWTTE

The initial activity of the source is 42 kBq. 33% of the alpha particles emitted by this source enter the chamber and form an ion pair.

Each alpha particle has an initial kinetic energy of 5.5 MeV.

The energy required to form one ion pair is 15 eV.

(c) Calculate the maximum current in the chamber due to the electrons when there is no smoke in the chamber. [3]

Markscheme

Each alpha gives rise to $\frac{5.5\times10^6}{15} = 3.67 \times 10^5$ ion pairs ✓

So $\frac{3.67\times10^5\times42000}{3} = 5.13 \times 10^9$ ion pairs per second ✓

current $= 1.6 \times 10^{-19} \times 5.13 \times 10^9 = 0.82 \times 10^{-9}$ «A» ✓

4. [Maximum mark: 8]

One possible fission reaction of uranium-235 $\left({}^{235}_{92}\mathrm{U}\right)$ is

$${}^{235}_{92}\mathrm{U} + n \rightarrow {}^{137}_{55}\mathrm{Cs} + {}^{95}_{37}\mathrm{Rb} + 4n$$

(a) Outline, with reference to the decay equation above, the role of chain reactions in the operation of a nuclear power station. [3]

Markscheme

The four neutrons released in the reaction may initiate further fissions ✓

«If sufficient U-235 is available,» the reaction is self-sustained ✓

Allowing for the continuous production of energy ✓

The number of neutrons available is controlled with control rods «to maintain the desired reaction rate» ✓

The following data are given:

Binding energy per nucleon of ${}^{235}_{92}\mathrm{U}$ = 7.591 MeV

Binding energy per nucleon of caesium-137 $\left({}^{137}_{55}\mathrm{Cs}\right)$ = 8.389 MeV

Binding energy per nucleon of rubidium-95 $\left({}^{95}_{37}\mathrm{Rb}\right)$ = 8.460 MeV

(b) Calculate, in MeV, the energy released in the reaction. [2]

Markscheme

$137 \times 8.389 + 95 \times 8.460 - 235 \times 7.591$ ✓

169 «MeV» ✓

(c) Two nuclides present in spent nuclear fuel $^{137}_{55}\text{Cs}$ are and cerium-144 ($^{144}_{58}\text{Ce}$). The initial activity of a sample of pure $^{144}_{58}\text{Ce}$ is about 40 times greater than the activity of the same amount of pure $^{137}_{55}\text{Cs}$.

Discuss which of the two nuclides is more likely to require long-term storage once removed from the reactor. [3]

Markscheme

«For the same number of nuclei,» the activity is inversely related to half-life ✓

Thus $^{137}_{55}\text{Cs}$ has a longer half-life and will likely require longer storage ✓

Half-lives of their decay products need also be considered when planning storage ✓

5. [Maximum mark: 7]

(a) Compare and contrast spontaneous and neutron-induced nuclear fission. [2]

Markscheme

Spontaneous fission occurs with no external influence, neutron-induced fission requires an interaction with a neutron «of appropriate energy» ✓

Both result in the release of energy

OR

both have a large number of possible pairs of products ✓

(b) Every neutron-induced fission reaction of uranium-235 releases an energy of about 200 MeV. A nuclear power station transfers an energy of about 2.4 GJ per second.

Determine the mass of uranium-235 that undergoes fission in one day in this power station. [3]

Markscheme

Fissions per day $= \frac{2.4\times10^{9}\times60\times60\times24}{200\times10^{6}\times1.6\times10^{-19}}$ «$= 6.5\times10^{24}$» ✓

Mass of uranium $= \frac{6.5\times10^{24}}{6.05\times10^{23}} \times 0.235$ ✓

2.5 «kg» ✓

(c) State **two** properties of the products of nuclear fission due to which the spent nuclear fuel needs to be kept safe. [2]

Markscheme

Have relatively short half-lives / high activity ✓

Their decay products are «usually» also radioactive ✓

Volatile / chemically active ✓

Biologically active / easily absorbed by living matter ✓

6. [Maximum mark: 1]

Which equation is an example of β^+ decay?

A. $^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + \mathrm{e}^- + \nu_\mathrm{e}$

B. $^{14}_{6}\mathrm{C} \rightarrow {}^{14}_{7}\mathrm{N} + \mathrm{e}^+ + \overline{\nu}_\mathrm{e}$

C. $^{23}_{12}\mathrm{Mg} \rightarrow {}^{23}_{11}\mathrm{Na} + \mathrm{e}^+ + \overline{\nu}_\mathrm{e}$

D. $^{23}_{12}\mathrm{Mg} \rightarrow {}^{23}_{11}\mathrm{Na} + \mathrm{e}^+ + \nu_\mathrm{e}$ [1]

Markscheme
D

7. [Maximum mark: 1]

The half-life of a radioactive nuclide is 15 minutes. The initial count rate observed from a pure source of the radioactive nuclide is 580 counts s^{-1}.

The average background count rate is 20 counts s^{-1}. The decay products are stable.

What is the observed count rate one hour after the initial observation?

A. 35 counts s^{-1}

B. 55 counts s^{-1}

C. 70 counts s^{-1}

D. 90 counts s^{-1} [1]

Markscheme

B

8. [Maximum mark: 1]

Which particle, at rest, has the largest magnitude for $\frac{\text{charge}}{\text{mass}}$?

A. Alpha particle

B. Beta minus particle

C. Proton

D. Neutron [1]

Markscheme

B

9. [Maximum mark: 1]

A nucleus of uranium undergoes fission.

$$^{1}_{0}\text{n} + {}^{235}_{92}\text{U} \rightarrow {}^{141}_{56}\text{Ba} + {}^{x}_{36}\text{Kr} + 3{}^{1}_{0}\text{n}$$

What is correct about the number of nucleons *x* in the Kr nucleus and the ratio

$$\frac{\text{binding energy per nucleon of Kr}}{\text{binding energy per nucleon of Ba}}?$$

	Number of nucleons x in Kr nucleus	$\frac{\text{binding energy per nucleon of Kr}}{\text{binding energy per nucleon of Ba}}$
A.	92	Greater than 1
B.	92	Less than 1
C.	94	Greater than 1
D.	94	Less than 1

[1]

Markscheme

A

10. [Maximum mark: 1]

The half-life of a radioactive substance is 5 hours. At time $t = 20$ hours N nuclei are present in the sample.

What was the number of radioactive nuclei that were present at time $t = 5$ hours?

A. $3N$

B. $4N$

C. $8N$

D. $9N$ [1]

Markscheme

C

11. [Maximum mark: 7]

(a) State what is meant by the half-life of a radioactive nuclide. [1]

Markscheme

** This question is from an examination for a previous syllabus. Please note that it may contain minor differences in marking or structure or contain parts that are not a match to the new syllabus (first assessment 2025).*

time for activity/number of « parent » nuclei in the sample to decrease to ½ of the initial value ✔

Allow 'number of radioactive particles'

DO NOT allow 'nuclide'

Tritium, $^{3}_{1}\mathrm{H}$, is a radioactive isotope of hydrogen. The activity of a sample of tritium decreases to $\frac{A_0}{8}$ after a time of 37.0 years where A_0 is the initial activity.

(b) Calculate the half-life of tritium. [2]

Markscheme

Alternative 1

«activity reduces to 1/8 after» 3 half-lives ✔

$T_{½}$=« $\frac{37.0}{3}$ =»12.3 «y» ✔

Alternative 2

$A=A_0e^{-\lambda t}$ to find λ, = 0.056 «y^{-1}» ✔

half life =« $\frac{\ln 2}{\lambda} = \frac{\ln 2}{0.056}$ =» 12.3 «y» ✔

Deuterium, ${}^{2}_{1}\mathrm{H}$, and tritium undergo nuclear fusion according to:

${}^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow \mathrm{X} + {}^{1}_{0}\mathrm{n}$

(c) Identify nuclide X. [1]

Markscheme

alpha particle / α / ${}^{4}_{2}\mathrm{He}$ ✔

The following data are given for binding energies per nucleon.

Nuclide	Binding energy per nucleon / MeV
${}^{2}_{1}\mathrm{H}$	1.112
${}^{3}_{1}\mathrm{H}$	2.827
X	7.074

(d.i) Show that the energy released in this reaction is about 18 MeV. [2]

Markscheme

$4 \times 7.074 - 2 \times 1.112 - 3 \times 2.827$ ✔

17.6 «MeV» ✔

MP1: Any one of the following calculations:

4 × 7.074

2 × 1.112

3 × 2.827

MP2: correct arithmetic, leading to 17.6 MeV

Must see full substitution or answer to at least 3sf for MP2

(d.ii) Outline why mass is not conserved in this reaction. [1]

Markscheme

some of the mass «of the reactants» converted to energy

«according to $\Delta m = \frac{\Delta E}{c^2}$» ✔

Need to see a connection between mass and energy.

Ignore subsequent description of energy (released, binding energy, etc.)

12. [Maximum mark: 8]

(a) Quarks are elementary particles.

(a.i) State what is meant by an elementary particle. [1]

Markscheme

** This question is from an examination for a previous syllabus. Please note that it may contain minor differences in marking or structure or contain parts that are not a match to the new syllabus (first assessment 2025).*

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

A particle without structure/constituents/component particles ✔

(a.ii) List the fundamental forces that act on quarks. [1]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

Electromagnetic, weak «nuclear», strong «nuclear» «and gravitational» ✔

Allow electroweak for electromagnetic and weak

(a.iii) The lambda baryon (Λ^0) has quark content uds. It decays according to the reaction $\Lambda^0 \rightarrow p + \pi^-$. The quark content of the pion is $\bar{u}d$.

State and explain which fundamental interaction is responsible for this decay. [2]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

Weak ✔

Decay violates strangeness «and only the weak interaction does» ✔

For MP2 allow comments that suggest that strangeness has changed in the interaction

(b.i) By reference to the concept of binding energy, explain why the combined mass of the nucleons of a nucleus is greater than the mass of the nucleus. [2]

Markscheme

According to $E = \Delta mc^2$ / identifies mass energy equivalence ✔

energy is released when nucleons come together / a nucleus is formed «so nucleus has less mass than individual nucleons»

OR

energy is required to «completely» separate the nucleons / break apart a nucleus «so individual nucleons have more mass than nucleus» ✔

(b.ii) The following binding energies per nucleon are available:

Th: 7.645074 MeV
Ra: 7.679917 MeV
He: 7.073915 MeV

Determine the energy released in the decay $^{228}_{90}\mathrm{Th} \rightarrow {}^{224}_{88}\mathrm{Ra} + {}^{4}_{2}\mathrm{He}$. [2]

Markscheme

$Q = 224 \times 7.679917 + 4 \times 7.073915 - 228 \times 7.645074$ ✔

$Q = 5.520$ «MeV» ✔

Allow 8.8×10^{-13} *J*

Award [1max] if answer is negative.

13. [Maximum mark: 1]

A sample of the artificial element Seaborgium-265 (Sg-265) is created with 4.0×10^{22} atoms. Sg-265 has a half-life of 15 s. How many atoms of Sg-265 remain after one minute?

A. 2.5×10^{21}

B. 5.0×10^{21}

C. 1.0×10^{22}

D. 2.0×10^{22} [1]

Markscheme
A

14. [Maximum mark: 1]

What is the number of protons and what is the number of neutrons in a nucleus of $^{214}_{82}\mathrm{Pb}$?

	Number of protons	Number of neutrons
A.	214	82
B.	82	214
C.	82	132
D.	132	214

[1]

Markscheme
C

15. [Maximum mark: 1]

A sample of the artificial element Seaborgium-265 (Sg-265) is created with 4.0×10^{22} atoms. Sg-265 has a half-life of 15 s. How many atoms of Sg-265 remain after one minute?

A. 2.5×10^{21}

B. 5.0×10^{21}

C. 1.0×10^{22}

D. 2.0×10^{22} [1]

Markscheme
A

16. [Maximum mark: 1]

What is the number of protons and what is the number of neutrons in a nucleus of $^{214}_{82}\mathrm{Pb}$?

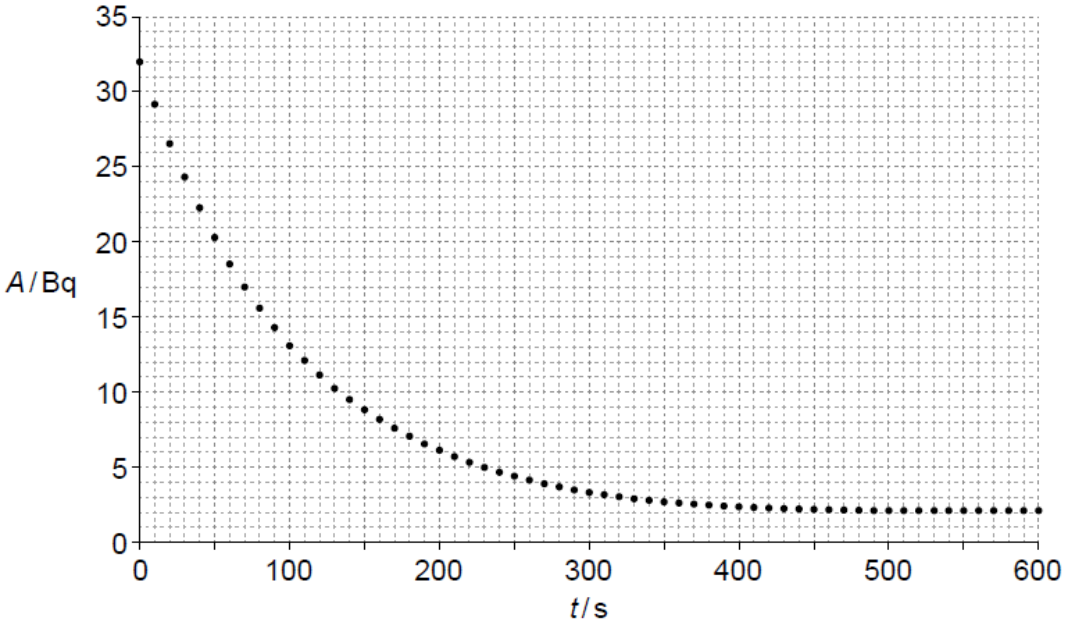
	Number of protons	Number of neutrons
A.	214	82
B.	82	214
C.	82	132
D.	132	214

[1]

Markscheme
C

17. [Maximum mark: 4]

The graph shows the variation with time t of the activity A of a sample of protactinium-234.



(a) Suggest why the activity approaches a non-zero constant value. [1]

Markscheme

there is background radiation present

OR

there is a systematic error in the rate counter ✔

Award [1] for an explanation that suggests that other radioactive elements of higher half-life could be present in the sample.

(b) Estimate the half-life of protactinium-234 explaining your work. [3]

Markscheme

realization that 2 «Bq» is the background activity ✔

calculation, reading or work on graph that shows understanding of half-life ✔

half-life = 70 (±5) «s» ✔

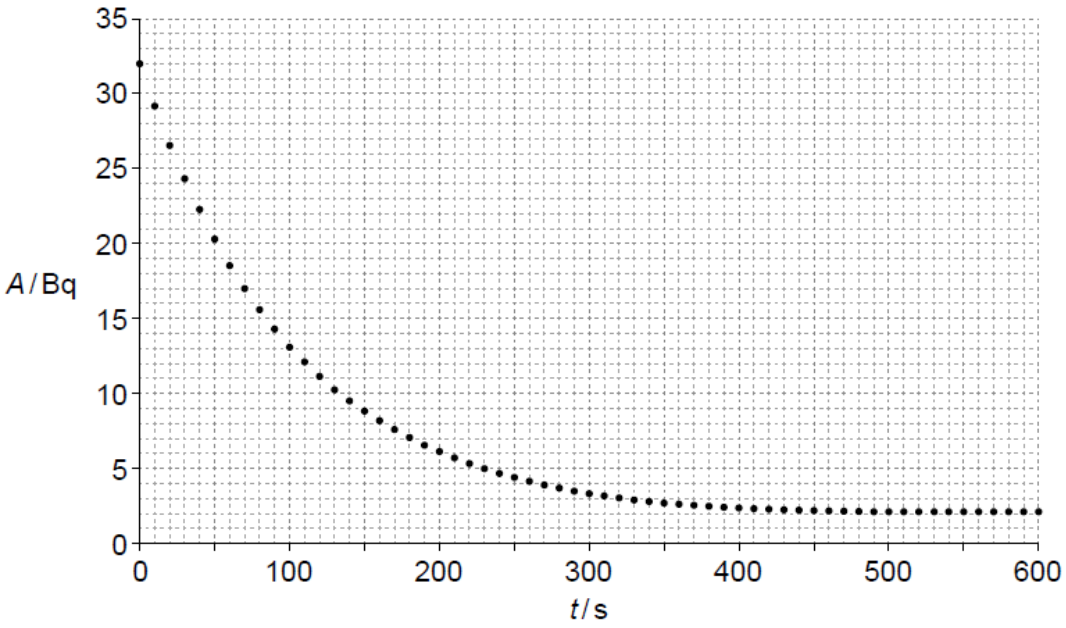
*Allow **[2] max** for a correct calculation of a half-life if MP1 is not scored, range accepted 65 to 85.*

*Award **MP2** for the correct use of any number of half-lives.*

*For **MP3**, accept answers in the range 65 «s» to 75 «s»*

18. [Maximum mark: 4]

The graph shows the variation with time *t* of the activity *A* of a sample of protactinium-234.



(a) Suggest why the activity approaches a non-zero constant value. [1]

Markscheme

there is background radiation present

OR

there is a systematic error in the rate counter ✔

Award [1] for an explanation that suggests that other radioactive elements of higher half-life could be present in the sample.

(b) Estimate the half-life of protactinium-234 explaining your work. [3]

Markscheme

realization that 2 «Bq» is the background activity ✔

calculation, reading or work on graph that shows understanding of half-life ✔

half-life = 70 (±5) «s» ✔

*Allow **[2] max** for a correct calculation of a half-life if MP1 is not scored, range accepted 65 to 85.*

*Award **MP2** for the correct use of any number of half-lives.*

*For **MP3**, accept answers in the range 65 «s» to 75 «s»*

19. [Maximum mark: 1]
The ratio of the diameter of an atom to the diameter of its nucleus is:

A. 10^1

B. 10^3

C. 10^5

D. 10^7 [1]

Markscheme
C

20. [Maximum mark: 1]
A radioactive material has a half-life of 6 days. How long will it take for 75 % of a pure sample of the material to decay?

A. 3 days

B. 6 days

C. 12 days

D. 18 days [1]

Markscheme
C

21. [Maximum mark: 1]
A student measures the count rate of a radioactive sample with time in a laboratory. The background count in the laboratory is 30 counts per second.

Count rate / counts per second	Time / s
150	0
90	20

What is the time at which the student measures a count rate of 45 counts per second?

A. 30 s

B. 40 s

C. 60 s

D. 80 s [1]

Markscheme
C

22. [Maximum mark: 1]
Three statements about the binding energy are provided.

I. The binding energy is the energy required to completely separate the nucleons.
II. The binding energy is equivalent, in units of energy, to the mass defect when a nucleus is formed from its nucleons.
III. The binding energy is the energy released when a nucleus is formed from its nucleons.

Which statements are true?

A. I and II only

B. I and III only

C. II and III only

D. I, II and III [1]

Markscheme
D

23. [Maximum mark: 1]

A nucleus of platinum (Pt) undergoes alpha decay to form an osmium (Os) nucleus as represented by the following reaction.

$^{175}_{78}\mathrm{Pt} \rightarrow$ Os + alpha particle

What are the number of protons and the number of neutrons in the osmium nucleus?

	Number of protons	Number of neutrons
A.	74	93
B.	76	93
C.	74	95
D.	76	95

[1]

Markscheme
D

24. [Maximum mark: 13]

(a) Identify with ticks [✓] in the table, the forces that can act on electrons and the forces that can act on quarks.

	Weak nuclear	Strong nuclear
Electrons		
Quarks		

[2]

Markscheme

Weak nuclear: 2 ticks ✓
Strong nuclear: quarks only ✓

(b) The following data is available for atomic masses for the fusion reaction

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}:$$

$^{2}_{1}\mathrm{H}$	2.0141 u
$^{3}_{1}\mathrm{H}$	3.0160 u
$^{4}_{2}\mathrm{He}$	4.0026 u

(b.i) Show that the energy released is about 18 MeV. [2]

Markscheme

«⊠» = 2.0141 + 3.0160 – (4.0026 + 1.008665) «= 0.0188 u»

OR

In MeV: 1876.13415 + 2809.404 – (3728.4219 + 939.5714475) ✓

= 0.0188 × 931.5 ***OR*** = 17.512 «MeV» ✓

*Must see either clear substitutions or answer to at least 3 s.f. for **MP2**.*

(b.ii) Estimate the specific energy of hydrogen by finding the energy produced when 0.4 kg of $^{2}_{1}\mathrm{H}$ and 0.6 kg of $^{3}_{1}\mathrm{H}$ undergo fusion. [2]

Markscheme

ALTERNATIVE 1

0.40 kg of deuterium is «$\frac{400}{2} \times 6.02 \times 10^{23}$» = 1.2×10^{26} nuclei
« 0.60 kg of tritium is the same number » ✓

So specific energy «$\frac{1.2\times10^{26}\times17.51\times10^{6}\times1.6\times10^{-19}}{0.4+0.6}$» = 3.4×10^{14} «J kg^{-1}»

ALTERNATIVE 2

«$17.51 \times 10^{6} \times 1.6 \times 10^{-19}$ =» 2.8×10^{-12} «J»

AND

«$(2.0141 + 3.0160) \times 1.66 \times 10^{-27}$ =» 8.35×10^{-27} ✓

«$\frac{2.8\times10^{-12}}{8.35\times10^{-37}}$» = 3.4×10^{14} «J kg^{-1}»

Allow $\sim 2.1 \times 10^{27}$ MeV kg^{-1} *for **MP2**.*

*Allow **ECF** from **MP1** for both **ALTs**.*

(c) It is hoped that nuclear fusion can be used for commercial production of energy.

Outline

(c.i) **two** difficulties of energy production by nuclear fusion. [2]

Markscheme

Requires high temp/pressure ✓
Must overcome Coulomb/intermolecular repulsion ✓
Difficult to contain / control «at high temp/pressure» ✓
Difficult to produce excess energy/often energy input greater than output / ***OWTTE*** ✓
Difficult to capture energy from fusion reactions ✓
Difficult to maintain/sustain a constant reaction rate ✓

(c.ii) **one** advantage of energy production by nuclear fusion compared to nuclear fission. [1]

Markscheme

Plentiful fuel supplies ***OR*** larger specific energy ***OR*** larger energy density ***OR*** little or no «major radioactive» waste products ✓

Allow descriptions such as "more energy per unit mass" or "more energy per unit volume"

(d) Tritium ($^{3}_{1}\mathrm{H}$) is unstable and decays into an isotope of helium (He) by beta minus decay with a half-life of 12.3 years.

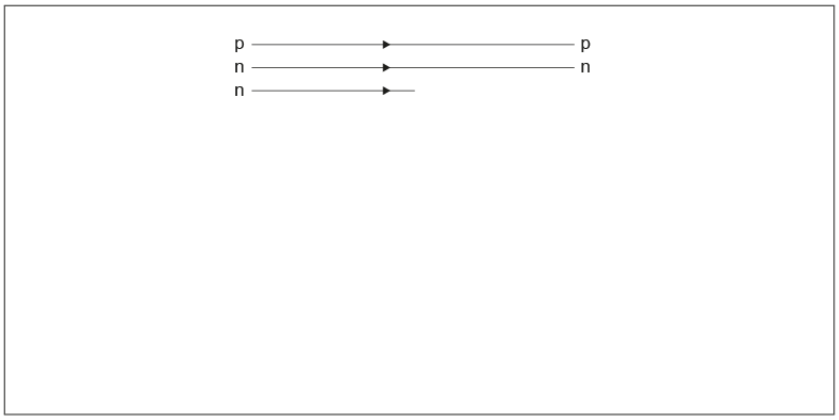
(d.i) State the nucleon number of the He isotope that $^{3}_{1}\mathrm{H}$ decays into. [1]

Markscheme

3 ✓

*Do **not** accept* $^{3}_{2}He$ *by itself.*

(d.ii) The following diagram is an incomplete Feynman diagram describing the beta minus decay of $^{3}_{1}\mathrm{H}$ into He. Complete the diagram and label all the missing particles.



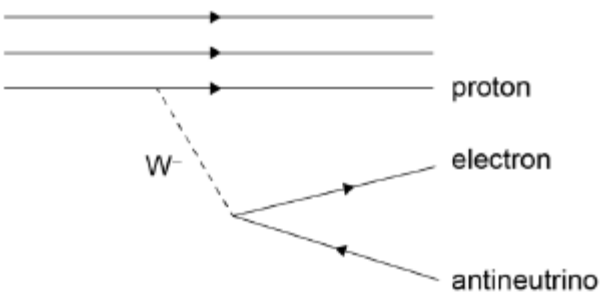
[3]

Markscheme

Proton shown ✓

W- shown ✓

Produces electron/e^- /$\boxtimes^-$ and antineutrino / $\overline{v}$ with proper arrow directions. ✓



Allow solid, dashed, or wavy line for W-particle.

Must see bar on antineutrino if symbol used.

25. [Maximum mark: 11]

Magnesium-27 nuclei ($^{27}_{12}\mathrm{Mg}$) decay by beta-minus (β^-) decay to form nuclei of aluminium-27 (Al).

(a) Show, using the data, that the energy released in the decay of one magnesium-27 nucleus is about 2.62 MeV.

Mass of aluminium-27 atom = 26.98153 u
Mass of magnesium-27 atom = 26.98434 u
The unified atomic mass unit is 931.5 MeV c^{-2}. [1]

Markscheme

(26.98434 - 26.98153) × 931.5

OR

2.6175 «MeV» seen ✓

(b) A Magnesium-27 nucleus can decay by one of two routes:

Route 1: 70% of the beta particles are emitted with a maximum kinetic energy of 1.76656 MeV, accompanied by a gamma photon of energy 0.84376 MeV.

Route 2: 30% of the beta particles have a maximum kinetic energy of 1.59587 MeV with a gamma photon of energy 1.01445 MeV.

The final state of the aluminium-27 nucleus is the same for both routes.

(b.i) State the conclusion that can be drawn from the existence of these two routes. [1]

Markscheme
evidence for nuclear energy levels ✓

(b.ii) Calculate the difference between the magnitudes of the total energy transfers in parts (a) and (b). [1]

Markscheme
Difference = 2.6175 – (1.76656 +0.84376) = 2.6175 – 2.61032 = 0.007195 «MeV» ***OR*** Difference = 2.6175 – (1.59587 +1.01445) = 2.61032 = 0.007195 «MeV» ✓

(b.iii) Explain how the difference in part (b)(ii) arises. [1]

Markscheme
Another particle/«anti» neutrino is emitted «that accounts for this mass / energy» ✓

(c) Small amounts of magnesium in a material can be detected by firing neutrons at magnesium-26 nuclei. This process is known as irradiation.

Magnesium-27 is formed because of irradiation. The products of the beta-particle emission are observed as the magnesium-27 decays to aluminium-27.

(c.i) The smallest mass of magnesium that can be detected with this technique is 1.1×10^{-8} kg.

Show that the smallest number of magnesium atoms that can be detected with this technique is about 10^{17}. [2]

Markscheme

So 1.1×10^{-8} kg $\equiv \frac{1.1}{0.027} \times 10^{-8}$ «mol»

OR

Mass of atom = $27 \times 1.66 \times 10^{-27}$ «kg» ✓

2.4 – 2.5 x 10^{17} atoms ✓

(c.ii) A sample of glass is irradiated with neutrons so that all the magnesium atoms become magnesium-27. The sample contains 9.50×10^{15} magnesium atoms.

The decay constant of magnesium-27 is 1.22×10^{-3} s^{-1}.

Determine the number of aluminium atoms that form in 10.0 minutes after the irradiation ends. [3]

Markscheme

$N_{10} = 9.50 \times 10^{15} \times e^{-0.00122 \times 60}$ seen ✓

$N_{10} = 4.57 \times 10^{15}$ ✓

So number of aluminium-27 nuclei = $(9.50 - 4.57) \times 10^{15} = 4.9(3) \times 10^{15}$ ✓

(c.iii) Estimate, in W, the average rate at which energy is transferred by the decay of magnesium-27 during the 10.0 minutes after the irradiation ends. [2]

Markscheme

Total energy released = ans **(c)(ii)** $\times 2.62 \times 10^{6} \times 1.6 \times 10^{-19}$ «= 2100 J» ✓

«$\frac{2100}{600}$ =» 3.4 –3.5 «W» ✓

26. [Maximum mark: 9]

When tritium ($^{3}_{1}\mathrm{H}$) decays by beta-minus (β^{-}) decay, one of the products is a stable isotope of helium (He).

(a) Outline what is meant by an isotope. [1]

Markscheme

«An atom with» the same number of protons ***AND*** different numbers of neutrons

OR

Same chemical properties ***AND*** different physical properties ✓

*Do **not** allow just atomic number and mass number*

(b) State, for the helium isotope produced in the tritium decay, its

(b.i) mass number. [1]

Markscheme

3 ✓

(b.ii) proton number. [1]

Markscheme

2 ✓

(c) Outline the quark change that occurs during this decay. [1]

Markscheme

d→u ✓

(d) Compare the properties of the strong nuclear force and of the electromagnetic force that allow the helium nucleus to be stable. [3]

Markscheme

Strong force is short range & electromagnetic force is long range ✓

Strong force is attractive between nucleons/neutrons & protons ✓

electromagnetic force is repulsive between protons ✓

Overall, the strong force dominates ✓

(e) A beta-minus particle and an alpha particle have the same initial kinetic energy.

Outline why the beta-minus particle can travel further in air than the alpha particle. [2]

Markscheme

Alphas have double charge «and so are better ionisers »✓

alphas have more mass and therefore slower «for same energy» ✓

so longer time/more likely to interact with the «atomic» electrons/atoms «and therefore better ionisers» ✓

Accept reverse argument in terms of betas travelling faster.

27. [Maximum mark: 1]
The unified atomic mass unit, u, is a non-SI unit usually used by scientists to state atomic masses.

What is u?

A. It is the mean of the masses of a proton and a neutron.

B. It is the mean of the masses of protons and neutrons in all chemical elements.

C. It is $\frac{1}{16}$ the mass of an $^{16}_{8}\mathrm{O}$ atom.

D. It is $\frac{1}{12}$ the mass of a $^{12}_{6}\mathrm{C}$ atom. [1]

Markscheme
D

28. [Maximum mark: 1]
The nuclide uranium-237 follows a sequence of three decays to produce the nuclide uranium-233.

What is a possible sequence for these decays?

A. Beta plus, alpha, beta plus

B. Beta minus, alpha, beta minus

C. Alpha, beta plus, beta minus

D. Alpha, beta minus, beta plus

[1]

Markscheme
B

29. [Maximum mark: 1]

A nucleus of krypton (Kr) decays to a nucleus of bromine (Br) according to the equation

$$^{84}_{36}\mathrm{Kr} \rightarrow {}^{84}_{35}\mathrm{Br} + \mathrm{Y} + \mathrm{Z}$$

What are Y and Z?

	Y	Z
A.	e^+	ν_e
B.	e^-	ν_e
C.	e^+	$\overline{\nu}_e$
D.	e^-	$\overline{\nu}_e$

[1]

Markscheme
A

30. [Maximum mark: 9]

Polonium-210 (Po-210) decays by alpha emission into lead-206 (Pb-206).

The following data are available.

Nuclear mass of Po-210 = 209.93676 u

Nuclear mass of Pb-206 = 205.92945 u

Mass of the alpha particle = 4.00151 u

(a) Outline, by reference to nuclear binding energy, why the mass of a nucleus is less than the sum of the masses of its constituent nucleons. [2]

Markscheme

according to $\Delta E = \Delta mc^2$ / identifies mass energy equivalence ✓

energy is released when nucleons come together / a nucleus is formed «so nucleus has less mass than individual nucleons»

OR

energy is required to «completely» separate the nucleons / break apart a nucleus «so individual nucleons have more mass than nucleus» ✓

Accept protons and neutrons.

(b.i) Calculate, in MeV, the energy released in this decay. [2]

Markscheme

$(m_{\text{polonium}} - m_{\text{lead}} - m_a)c^2$ ***OR*** (209.93676 – 205.92945 – 4.00151)

OR

mass difference = 5.8×10^{-3} ✓

conversion to MeV using 931.5 to give 5.4 «MeV» ✓

*Allow **ECF** from **MP1**.*

*Award **[2]** for a **BCA**.*

*Award **[1]** for 8.6 x 10^{-13} J.*

(b.ii) The polonium nucleus was stationary before the decay.

Show, by reference to the momentum of the particles, that the kinetic energy of the alpha particle is much greater than the kinetic energy of the lead nucleus. [3]

Markscheme

ALTERNATIVE 1

energy ratio expressed in terms of momentum, e.g. $\frac{E_\alpha}{E_{\text{lead}}} = \frac{p_\alpha^2/2m_\alpha}{p_{\text{lead}}^2/2m_{\text{lead}}}$ ✓

$p_\alpha = p_{\text{lead}}$ hence $\frac{E_\alpha}{E_{\text{lead}}} = \frac{m_{\text{lead}}}{m_\alpha}$ ✓

$\frac{m_{\text{lead}}}{m_\alpha} \simeq \frac{206}{4} = 51.5 \Rightarrow E_\alpha = 51.5E_{\text{lead}}$ «so α has a much greater KE»

OR

m_{lead} «much» greater than $\mathrm{m}_{\text{alpha}}$ «so α has a much greater KE» ✓

ALTERNATIVE 2

alpha particle and lead particle have equal and opposite momenta ✓

so their velocities are inversely proportional to mass ✓

but KE $\propto v^2$ «so α has a much greater KE» ✓

(b.iii) In the decay of polonium-210, alpha emission can be followed by the emission of a gamma photon.

State and explain whether the alpha particle or gamma photon will cause greater ionization in the surrounding material. [2]

Markscheme

alpha particle ✓

is electrically charged hence more likely to interact with electrons «in the surrounding material» ✓

31. [Maximum mark: 1]

A pure sample of iodine-131 decays into xenon with a half-life of 8 days.

What is $\frac{\text{number of iodine atoms remaining}}{\text{number of xenon atoms formed}}$ after 24 days?

A. $\frac{1}{8}$

B. $\frac{1}{7}$

C. $\frac{7}{8}$

D. $\frac{8}{7}$ [1]

Markscheme
B

32. [Maximum mark: 1]

The background count in a laboratory is 20 counts per second. The initial observed count rate of a pure sample of nitrogen-13 in this laboratory is 180 counts per second. The half-life of nitrogen-13 is 10 minutes. What is the expected count rate of the sample after 30 minutes?

A. 20 counts per second

B. 23 counts per second

C. 40 counts per second

D. 60 counts per second [1]

Markscheme
C

33. [Maximum mark: 1]

$^{238}_{92}\mathrm{U}$ undergoes an alpha decay, followed by a beta-minus decay.

What is the number of protons and neutrons in the resulting nuclide?

[1]

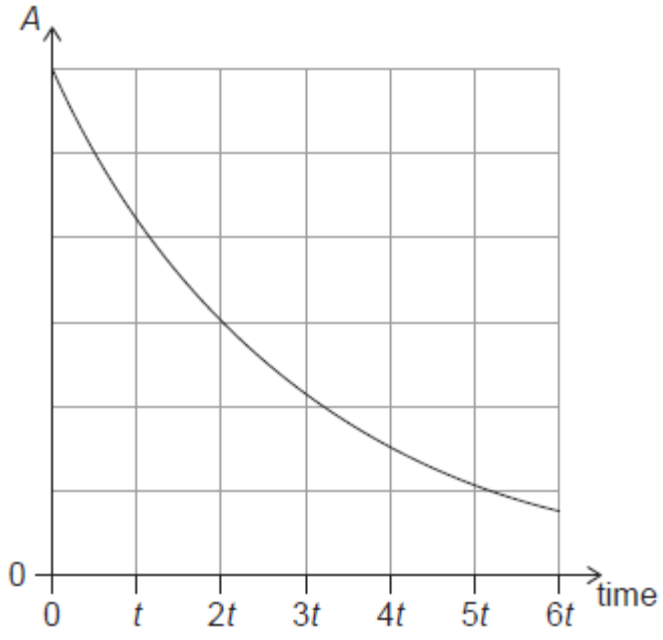
	Number of protons	Number of neutrons
A.	89	234
B.	91	144
C.	91	143
D.	89	145

Markscheme

C

34. [Maximum mark: 1]

A radioactive nuclide X decays into a nuclide Y. The graph shows the variation with time of the activity A of X. X and Y have the same nucleon number.



What is true about nuclide X?

A. alpha (α) emitter with a half-life of t

B. alpha (α) emitter with a half-life of 2*t*

C. beta-minus (β^-) emitter with a half-life of *t*

D. beta-minus (β^-) emitter with a half-life of 2*t* [1]

Markscheme
D

35. [Maximum mark: 1]
What statement is **not** true about radioactive decay?

A. The percentage of radioactive nuclei of an isotope in a sample of that isotope after 7 half-lives is smaller than 1 %.

B. The half-life of a radioactive isotope is the time taken for half the nuclei in a sample of that isotope to decay.

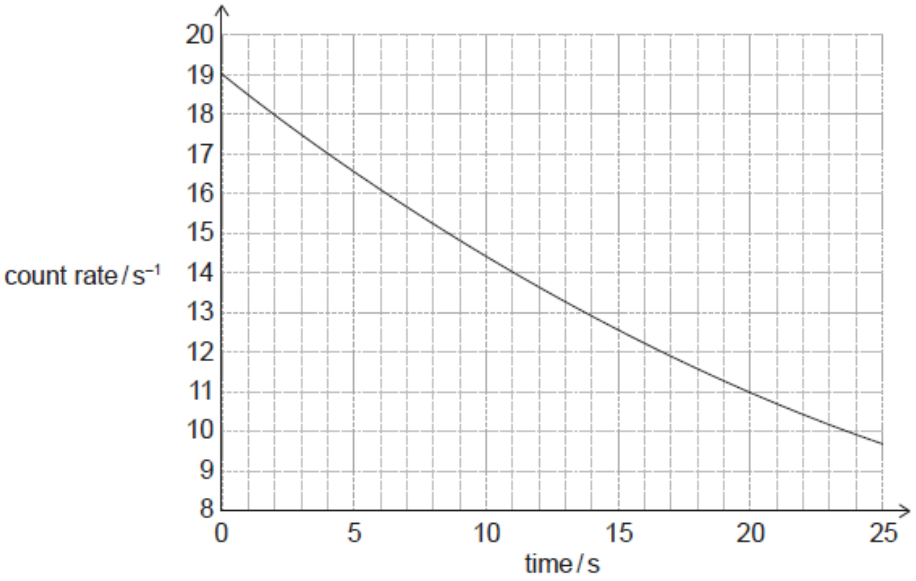
C. The whole-life of a radioactive isotope is the time taken for all the nuclei in a sample of that isotope to decay.

D. The half-life of radioactive isotopes range between extremely short intervals to thousands of millions of years. [1]

Markscheme
C

36. [Maximum mark: 1]
A detector measures the count rate from a sample of a radioactive nuclide. The graph shows the variation with time of the count rate.

The nuclide has a half-life of 20 s. The average background count rate is constant.



What is the average background count rate?

A. 1 s^{-1}

B. 2 s^{-1}

C. 3 s^{-1}

D. 4 s^{-1} [1]

Markscheme
C

37. [Maximum mark: 1]

A pure sample of radioactive nuclide X decays into a stable nuclide Y.

What is $\frac{\text{number of atoms of Y}}{\text{number of atoms of X}}$ after two half-lives?

A. 1

B. 2

C. 3

D. 4 [1]

Markscheme
C

38. [Maximum mark: 1]

The mass of a nucleus of iron-56 ($^{56}_{26}\mathrm{Fe}$) is M.

What is the mass defect of the nucleus of iron-56?

A. $M - 26m_p - 56m_n$

B. $26m_p + 30m_n - M$

C. $M - 26m_p - 56m_n - 26m_e$

D. $26m_p + 30m_n + 26m_e - M$ [1]

Markscheme
B

39. [Maximum mark: 1]

What is the relation between the value of the unified atomic mass unit in grams and the value of Avogadro's constant in mol^{-1}?

A. Their ratio is 1.

B. Their product is 1.

C. Their sum is 1.

D. Their difference is 0. [1]

Markscheme
B

40. [Maximum mark: 1]

Three particles are produced when the nuclide $^{23}_{12}\mathrm{Mg}$ undergoes beta-plus (β^+) decay. What are two of these particles?

A. $^{23}_{11}\mathrm{Na}$ and $^{0}_{0}\mathrm{v_e}$

B. $^{0}_{-1}\mathrm{e}$ and $^{0}_{0}\mathrm{v_e}$

C. $^{23}_{11}\mathrm{Na}$ and $^{0}_{0}\overline{\mathrm{v}}_{\mathrm{e}}$

D. $^{0}_{1}\mathrm{e}$ and $^{0}_{0}\overline{\mathrm{v}}_{\mathrm{e}}$ [1]

Markscheme
A

41. [Maximum mark: 1]

When a high-energy α-particle collides with a beryllium-9 ($^{9}_{4}\mathrm{Be}$) nucleus, a nucleus of carbon $(\mathrm{Z} = 6)$ may be produced. What are the

products of this reaction?

	Product 1	Product 2
A.	carbon-12	proton
B.	carbon-12	neutron
C.	carbon-14	proton
D.	carbon-14	neutron

[1]

Markscheme
B

42. [Maximum mark: 1]

The mass of nuclear fuel in a nuclear reactor decreases at the rate of $8\ \text{mg}$ every hour. The overall reaction process has an efficiency of $50\,\%$. What is the maximum power output of the reactor?

A. $100\ \text{MW}$

B. $200\ \text{MW}$

C. $100\ \text{GW}$

D. $200\ \text{GW}$

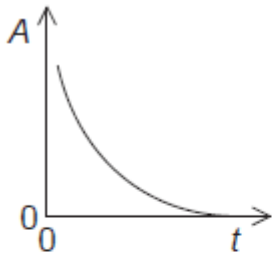
[1]

Markscheme
A

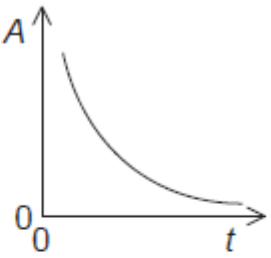
43. [Maximum mark: 1]

Which graph shows the variation of activity A with time t for a radioactive nuclide?

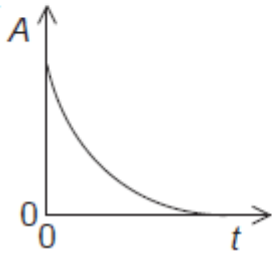
A.



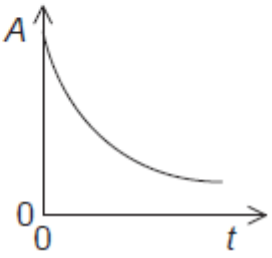
B.



C.



D.



[1]

Markscheme
D

44. [Maximum mark: 1]

What statement about alpha particles, beta particles and gamma radiation is true?

A. Gamma radiation always travels faster than beta particles in a vacuum.

B. In air, beta particles produce more ions per unit length travelled than alpha particles.

C. Alpha particles are always emitted when beta particles are emitted.

D. Alpha particles are deflected in the same direction as beta particles in a magnetic field. [1]

Markscheme
A

45. [Maximum mark: 1]

Nuclide X can decay by two routes. In Route 1 alpha (α) decay is followed by beta-minus (β^-) decay. In Route 2 β^- decay is followed by α decay. P and R are the intermediate products and Q and S are the final products.

Route 1 $\quad X \xrightarrow{\alpha} P \xrightarrow{\beta^-} Q$

Route 2 $\quad X \xrightarrow{\beta^-} R \xrightarrow{\alpha} S$

Which statement is correct?

A. Q and S are different isotopes of the same element.

B. The mass numbers of X and R are the same.

C. The atomic numbers of P and R are the same.

D. X and R are different isotopes of the same element. [1]

Markscheme
B

46. [Maximum mark: 1]

Gamma (γ) radiation

A. is deflected by a magnetic field.

B. affects a photographic plate.

C. originates in the electron cloud outside a nucleus.

D. is deflected by an electric field. [1]

Markscheme
B

47. [Maximum mark: 1]

A proton, an electron and an alpha particle are at rest. Which particle has the smallest magnitude of ratio of charge to mass and which particle has the largest magnitude of ratio of charge to mass?

	Smallest charge to mass ratio	Largest charge to mass ratio
A.	alpha	electron
B.	electron	alpha
C.	electron	proton
D.	proton	electron

[1]

Markscheme
A

48. [Maximum mark: 1]

X is a radioactive nuclide that decays to a stable nuclide. The activity of X falls to $\frac{1}{16}$ th of its original value in 32 s.

What is the half-life of X?

A. 2 s

B. 4 s

C. 8 s

D. 16 s [1]

Markscheme
C

49. [Maximum mark: 7]

A stationary nucleus of uranium-238 undergoes alpha decay to form thorium-234.

The following data are available.

Energy released in decay	4.27 MeV
Binding energy per nucleon for helium	7.07 MeV
Binding energy per nucleon for thorium	7.60 MeV

(a) Radioactive decay is said to be "random" and "spontaneous". Outline what is meant by each of these terms.

Random:

Spontaneous: [2]

Markscheme

random:

it cannot be predicted which nucleus will decay

OR

it cannot be predicted when a nucleus will decay ✔

NOTE: OWTTE

spontaneous:

the decay cannot be influenced/modified in any way ✔

NOTE: OWTTE

(b(i)) Calculate the binding energy per nucleon for uranium-238. [3]

Markscheme

234×7.6 ***OR*** 4×7.07 ✔

BE_U =« 234 × 7.6 + 4 × 7.07 – 4.27 =» 1802 « MeV » ✔

$\frac{BE_U}{A} = « \frac{1802}{238} = » 7.57$ « MeV » ✔

NOTE: Allow ECF from MP2

*Award **[3]** for bald correct answer*

Allow conversion to J, final answer is 1.2 × 10^{-12}

(b(ii)) Calculate the ratio $\frac{\text{kinetic energy of alpha particle}}{\text{kinetic energy of thorium nucleus}}$. [2]

Markscheme

states or applies conservation of momentum ✔

ratio is « $\frac{E_{k\alpha}}{E_{kTh}} = \frac{\frac{p^2}{2m_\alpha}}{\frac{p^2}{2m_{Th}}} = \frac{234}{4}$ » 58.5 ✔

*NOTE: Award **[2]** for bald correct answer*

50. [Maximum mark: 1]

The carbon isotope $^{14}_{6}C$ is radioactive. It decays according to the equation

$$^{14}_{6}C \rightarrow {}^{14}_{7}N + X + Y$$

What are X and Y?

	X	Y
A.	$^{0}_{-1}\beta$	ν_e
B.	$^{0}_{-1}\beta$	$\bar{\nu}_e$
C.	$^{0}_{1}\beta$	ν_e
D.	$^{0}_{1}\beta$	$\bar{\nu}_e$

[1]

Markscheme
B

51. [Maximum mark: 1]

Which property of a nuclide does **not** change as a result of beta decay?

A. Nucleon number

B. Neutron number

C. Proton number

D. Charge

[1]

Markscheme
A

52. [Maximum mark: 1]

The rest mass of the helium isotope $^{3}_{2}\mathrm{He}$ is *m*.

Which expression gives the binding energy per nucleon for $^{3}_{2}\mathrm{He}$?

A. $\frac{(2m_p+m_n+m)c^2}{3}$

B. $\frac{(2m_p+m_n-m)c^2}{3}$

C. $(2m_p + m_n + m)c^2$

D. $(2m_p + m_n - m)c^2$

[1]

Markscheme
B

53. [Maximum mark: 1]

A radioactive nuclide with atomic number *Z* undergoes a process of beta-plus (β^+) decay. What is the atomic number for the nuclide produced and what is another particle emitted during the decay?

	Atomic number	Particle
A.	Z – 1	neutrino
B.	Z + 1	neutrino
C.	Z – 1	anti-neutrino
D.	Z + 1	anti-neutrino

[1]

Markscheme

A

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